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Bellcore GR-326-CORE
Issue 1, December 1994
Section 4.4.2.8 - Dust

Objectives:

O4-54 [72] No connector assembly should have a Loss exceeding 0.3 dB.

O4-55 [73] No connector assembly should have a Loss Increase exceeding 0.2 dB.

Conditional Requirement:

CR4-56 [74] No connector assembly shall have a Reflectance exceeding -55 dB.

Conditional Objective:

CO4-57 [75] No connector assembly should have a Reflectance exceeding -65 dB.

4.4.2.8 Dust

This test is conducted with Type I samples (Section 4.1). Before the samples are subjected to the dust environment, the samples for this test may be screened on the basis of loss and reflectance (unless otherwise specified by the test program under which the tests are being conducted).

During the Dust Test, the connector assemblies are exposed to an environment containing a heavy concentration of coarse dust. The connector assemblies are exposed both as mated assemblies and with one plug removed from the assembly. After exposure to the dust condition, loss and reflectance are measured. The connector assemblies are disconnected, cleaned, and reconnected; and more measurements are made.

The results of this test indicate the performance of connector assemblies after exposure to severe dust conditions.

Ten new connector assemblies are subjected to the Dust Test. Five of the connector assemblies, Set A, are to have both connector plugs mated during dust exposure, and the other 5, Set B, are to have one plug in each of the assemblies disconnected. If dust caps and covers are supplied with the adapters or plugs, the disconnected plugs and the open adapters are covered or capped during the Dust Test.

The test facility, dust material, and procedures for conducting the Dust Test are described in FOTP-35. Dust Test procedures are as follows:

- a. Clean the plugs and adapters per Cleaning Procedure "B" in Section 4.3.2, and assemble the connector assemblies.
- b. Measure Loss and Reflectance.

- c. Disconnect one plug from each of the assemblies in Set B and cap the respective plugs and adapters.
- d. Subject all the samples to the Dust Test as described in FOTP-35.
- e. Remove the samples from the dust chamber and, without disconnecting any of the plugs or removing any of the dust caps, clean the exposed surfaces of the connector assemblies and the sections of cable exposed to the dust. Cleaning may be done by: brushing, wiping (with a dry and/or wet wipe), and/or blowing with canned compressed gas.
- f. Remove the dust caps from the adapters and plugs in Set B. Clean the plugs and the adapter ends that were capped, and reconnect the plugs to the respective adapters. Use the Cleaning Procedure "A" as described in Section 4.3.1.
- g. Measure Loss and Reflectance for all the connector assemblies.
- h. Clean all connector assemblies in the Dust Test. Remove one plug from each adapter and using Cleaning Procedure "A" in Section 4.3.1. Always remove the same plug from each connector assembly.
- i. Repeat steps "i" and "j."
- j. Clean all connector assemblies using Cleaning Procedure "B" (Section 4.3.2).
- k. Measure Loss and Reflectance.

Criteria are applied to the Loss Increase (with respect to the PreDust measurement) and Reflectance measurements made after exposure to dust:

Requirements:

R4-58 [76] No connector assembly shall have a Loss Increase exceeding 0.3 dB.

R4-59 [77] No connector assembly shall have a Reflectance exceeding -40 dB.

Objective:

O4-60 [78] No connector assembly should have a Loss Increase exceeding 0.2 dB.

Conditional Requirement:

CR4-61 [79] No connector assembly shall have a Reflectance exceeding -55 dB.

Conditional Objective:

CO4-62 [80] No connector assembly should have a Reflectance exceeding -65 dB.

Table 4-1. List of Connector Tests

TESTS	CRITERIA	SECTION
2. Ferrule End Geometry	$10 \leq \text{radius} \leq 30 \text{ mm}$	4.4.5.2
3. Connector Installation	Loss Increase $\leq 0.1 \text{ dB}$	4.4.5.3

Measurements of optical performance, i.e., Loss and Reflectance, are made with a facility meeting the requirements in Section 5.2. Facilities capable of performing these measurements are described in Sections 5.2.1 and 5.2.2.

All measurements of optical performance are made at both 1310 nm and at 1550 nm, unless otherwise specified.

4.1 Test Samples

There are two types of Test Samples:

- **Type I Test Sample**, a connector assembly with unterminated leads. The sample consists of two plugs with attached leads, and an adapter. The 2 plugs with the attached leads may be provided for the tests as a jumper cable assembly.


Figure 4-1. Type I Test Sample

- **Type II Test Sample**, a jumper cable assembly with plugs and adapters to form connector assemblies with unterminated leads on either end. The sample consists of a jumper cable at least 2 meters long, two connector plugs with attached leads, and two adapters. The 2 plugs with the attached leads may be provided for the tests as a jumper cable assembly.


Figure 4-2. Type II Test Sample

The size of the sample of product and the type of samples that are subjected to the test procedures in this section are found in Sections 6, and 7. For tests not conducted under either of these sections, the user may determine the size of the sample.

FOTP-35
Fiber Optic Component Dust (Fine Sand) Test
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FOTP-35

Fiber Optic Component Dust (Fine Sand) Test

EIA/TIA-455-35A

(Revision of EIA-455-35)

MAY 1990

**ELECTRONIC INDUSTRIES ASSOCIATION
ENGINEERING DEPARTMENT**



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FOTP-35

FIBER OPTIC COMPONENT DUST (FINE SAND) TEST

(From EIA Standards Proposal No. 1894, formulated under the cognizance of TIA FO-6.3, Subcommittee on Fiber Optic Interconnecting Devices, and TIA FO-6.4, Subcommittee on Fiber Optic Test Methods and Instrumentation.)

This FOTP is part of the series of test procedures included within Recommended Standard EIA/TIA-455.

NOTE: This FOTP was previously published in RS-455-35 as FOTP-35.

1.0 INTENT

- 1.1 The dust test is used to ascertain the ability of fiber optic components to resist the effects of a dry dust (fine sand) laden atmosphere. This test simulates the effect of sharp edged dust (fine sand) particles, up to 150 μm in size, which may penetrate into cracks, crevices and joints. This test is applicable to all optical devices and to combinations of optical devices and mechanical, electrical, electronic, electrochemical, or electromechanical devices for which exposure to the effects of a dry dust (fine sand) laden atmosphere is anticipated.
- 1.2 General effects resulting from the penetration of dust can include a variety of damage such as fouling of moving parts, formation of electrically conductive bridges with resulting shorts, scratching of the optical surfaces of the fiber interfaces, and acting as a nucleus for collection of water vapor and, hence, a source of possible corrosion and malfunction of equipment.

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2.0 REFERENCED DOCUMENTS

Test or inspection requirements may include, but are not limited to, the following references:

FOTP-13 (EIA-455-13)

"Visual and Mechanical Inspection
of Fiber, Cables, Connectors
and/or Other Fiber Optic Devices"

3.0 TEST EQUIPMENT

3.1 Test Facility

The test facility shall consist of a chamber and accessories to control dust concentration and the velocity, temperature and humidity of dust-laden air. In order to provide adequate circulation of the dust-laden air, no more than 50% of the cross-sectional area (normal to the air flow) and no more than 30% of the volume of the chamber shall be occupied by the test item(s). The chamber shall be provided with a suitable means of maintaining and verifying the dust concentration in circulation. A minimum acceptable means for doing this is by use of a properly calibrated smoke meter and a standard light source. The dust laden air shall be introduced into the test space in such a manner as to allow it to become approximately laminar in flow before it strikes the test item.

3.2 Dust Requirements

The dust used in this test shall be fine sand (97%-99% by weight of SiO_2) of angular structure, and shall have the following size distribution, as determined by weight, using the U.S. Standard Sieve Series:

- (a) 100% of the dust shall pass through a 100-mesh screen,
- (b) $98 \pm 2\%$ of the dust shall pass through a 140-mesh screen,
- (c) $90 \pm 2\%$ of the dust shall pass through a 200-mesh screen, and
- (d) $75 \pm 2\%$ of the dust shall pass through a 325-mesh screen.

NOTE: 140-mesh silica flour as produced by the Ottawa Silica Company, Ottawa, IL, or equivalent, is satisfactory for use in the performance of these tests.

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4.0 TEST SPECIMEN

Unless otherwise specified:

- 4.1 A test specimen shall consist of a designated fiber optic component, prepared as intended for normal service, complete with accessories.
- 4.2 Connector specimens (or other components which incorporate connectors) shall consist of a mated connector pair, terminated as intended for normal service, and complete with accessories.
- 4.3 The combined length of any device with attached cable shall be 2.0 ± 0.1 m (6.6 ± 0.3 ft).

5.0 TEST PROCEDURE

- 5.1 Prepare the test specimen in accordance with Clause 4.0 and as may be directed by the Detail Specification.
- 5.2 Unless otherwise specified, test item(s) shall be subjected to the following procedures while in the non-operating mode. If an operating-mode test is required, operating and measurement instructions shall be provided by the Detail Specification.
- 5.3 Position the test item(s) as near to the center of the test chamber as practicable. If more than one item is being tested, there shall be a minimum clearance of 100 mm (4 in) between surfaces of the test item(s) and any other material or object capable of furnishing protection. Also, no surface of the test item(s) shall be closer than 100 mm (4 in) from any wall of the test chamber. Orient the item(s) so as to expose the most critical or vulnerable parts to the dust stream. The test item orientation may be changed during the test if required by the Detail Specification.
- 5.4 Set the chamber controls to maintain an internal chamber temperature of $23 \pm 2^{\circ}\text{C}$ (approx. 73°F) and a relative humidity of less than 22%. Adjust the air velocity to 533 ± 76 m per minute ($1,750 \pm 250$ ft per minute). Adjust the dust feeder to control the dust concentration at 10.6 ± 7.1 grams per cubic meter. Unless otherwise specified, maintain these chamber conditions for 6 hours.

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- 5.5 Upon completion of the 6 hour conditioning, stop the dust feed and reduce the air velocity to 91 ± 61 m per minute (300 ± 200 ft per minute). Raise the internal chamber temperature to $63 \pm 2^{\circ}\text{C}$ (approx. 145°F). Maintain a relative humidity of less than 10%. Unless otherwise specified, maintain these chamber conditions for 16 hours. 4 hrs temp
16 hrs test
- 5.6 Unless otherwise specified:
- (a) Upon completion of the 16 hour conditioning, maintain the chamber temperature at $63 \pm 2^{\circ}\text{C}$ (approx. 145°F).
 - (b) Adjust the air velocity to 533 ± 76 m per minute ($1,750 \pm 250$ ft per minute).
 - (c) Adjust the dust feeder to control the dust concentration at 10.6 ± 7.1 grams per cubic meter.
 - (d) Maintain a relative humidity of less than 10%. 6 hrs
 - (e) Maintain these chamber conditions for 6 hours.
- 5.7 Turn off all chamber controls and allow the test item(s) to return to standard ambient conditions. Remove accumulated dust from the test item(s) by brushing, wiping or shaking, exercising care to avoid introducing any additional dust into the test item(s). Dust shall not be removed by either air blast or vacuum cleaning.
- 5.8 In accordance with the requirements of FOTP-13, examine the test item(s) and record observations as specified in the Detail Specification. In the performance of this inspection, carefully examine test items containing bearings, seals, lubricants, etc., for the presence of dust deposits. Prior to remating components as specified in FOTP-13, normal cleaning procedures may be employed to thoroughly remove any residual dust.

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6.0 DOCUMENTATION

Test data sheets shall contain the following information:

- 6.1 Date and title of the test.
- 6.2 Identification of the fiber optic device (connector or other), attached fiber optic cable, and any other applicable data.
- 6.3 Test Equipment and date of calibration.
- 6.4 Identification of test methods or procedures used.
- 6.5 Results of specified examinations.
- 6.6 Names of test personnel.

7.0 SPECIFYING INFORMATION

The following details shall be specified in the Detail Specification:

- 7.1 Pre-test and post-test data required.
- 7.2 Failure or acceptance criteria.
- 7.3 Change in orientation during test, if required. (See 5.3.)
- 7.4 Whether equipment is to operate during test, and the length of time required for operation and measurements. (See 5.2, 5.4, 5.5 and 5.6.)
- 7.5 Whether the second 6-hour test at 63°C (145°F) shall be performed immediately after reaching stabilization. (See 5.6.)
- 7.7 Whether connectors are mated or unmated. (See 4.2.)
- 7.8 Overall specimen length, if other than 2 m (6.6 ft). (See 4.3.)

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Issue 2, December 1996
Section 4.4.4.1/5.3.7 – Dust Test

4.4.4 Materials and Environmental Tests

4.4.4.1 Dust Test

The Dust Test of GR-326-CORE, Issue 1 used a coarse dust which is not representative of the dust expected to be most troublesome for connectors. This revised Dust Test utilizes a fine dust known as "Arizona Road Dust." The results of this test indicate the performance of connector assemblies after exposure to severe dust conditions. The test can be performed in the facility described in Section .

Ten new connector assemblies are subjected to the Dust Test. Five of the connector assemblies, samples #1-5, are to have both connector plugs mated during dust exposure (Set "A", Figure 5-9A), and the other five, samples #6-10, are to have one plug in each of the assemblies disconnected (Set "B", Figure 5-9B). If dust caps and covers are supplied with the adapters or plugs, the disconnected plugs and the open adapters are covered or capped during the Dust Test. Details are given in Section . Samples may be pre-screened with respect to loss and reflectance, to ensure that the initial population consists of conforming product. The cleaning procedure to be used throughout is cleaning Procedure "A" in Section 4.3.1 or cleaning Procedure "B" in Section 4.3.2, at the supplier's option.

The Dust Test procedure is as follows:

- a. Clean the plugs and adapters and assemble the connector assemblies.
 - b. Measure loss and reflectance.
 - c. Disconnect one plug (plug a) from each of the assemblies in samples #6-10 and cap the respective plugs and adapters (see Figure 5-9B).
 - d. Subject all the samples to the Dust Test as described in Section .
 - e. Remove the samples from the dust chamber and, without disconnecting any of the plugs or removing any of the dust caps, clean the exposed surfaces of the connector assemblies and the sections of cable exposed to the dust. Cleaning may be done by: brushing, wiping (with a dry and/or wet wipe), and blowing with canned compressed gas.
 - f. Measure loss and reflectance of only the samples that were mated during dust exposure, samples #1-5.
 - g. Disconnect one plug (plug a) from samples #1-5. Leave the remaining plug (plug b) as a bulkhead configuration, i.e. still connected to the adapter. Clean all disconnected plugs and all bulkhead configurations (see Figure 5-9B).
 - h. Re-connect and measure loss and reflectance for all ten connector assemblies.
-

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- i. Repeat steps "g" and "h" two additional times. The plugs still inside an adaptor may be cleaned with canned air, alcohol and a cotton swab. Always remove the same plug from each connector assembly.
- j. Disconnect all plugs from the adaptors, clean thoroughly with procedures appropriate to plugs and adaptors respectively.
- k. Re-connect all plugs and adaptors as before, measure loss and reflectance for all ten connector assemblies.
- l. Repeat steps "j" and "k" two additional times.

These steps are summarized in a flow chart in Figure 4-6.

Criteria are applied to the changes in loss and reflectance measurements made after exposure to dust compared to the Pre-Dust measurement.

R4-44 [62]Dust Test Requirements: Set "A"

- ... The product of Set "A" (Group I) shall meet the loss and reflectance Requirements criteria listed in Tables 4-2 and 4-3 in the row marked "During Test, Not Under Load" during the first measurement (before cleaning).

RATIONALE — At this point the product will not have been demated, so its performance should not be affected by dust exposure.

R4-45 [63]Dust Test Requirements: Sets "A" and "B"

- ... The product of both Sets "A" and "B" (Groups II and III) shall meet the loss and reflectance Requirements criteria listed in Tables 4-2 and 4-3 in the row marked "During Test, Not Under Load" after the last cleaning in Step "l."

RATIONALE — At this point the product will have been thoroughly cleaned three times, so performance should be returned to conforming levels.

O4-46 [64]Dust Test Objectives

- ... The product of both Sets "A" and "B" (Groups II and III) should meet the loss and reflectance **Requirements** (not Objectives) criteria listed in Tables 4-2 and 4-3 in the row marked "During Test, Not Under Load" after the last cleaning in Step "i."

RATIONALE — At this point the product will have been cleaned three times but with one plug still in the adaptor; i.e., not as thoroughly as possible.

O4-47 [65]Dust Test Cleanability Objective

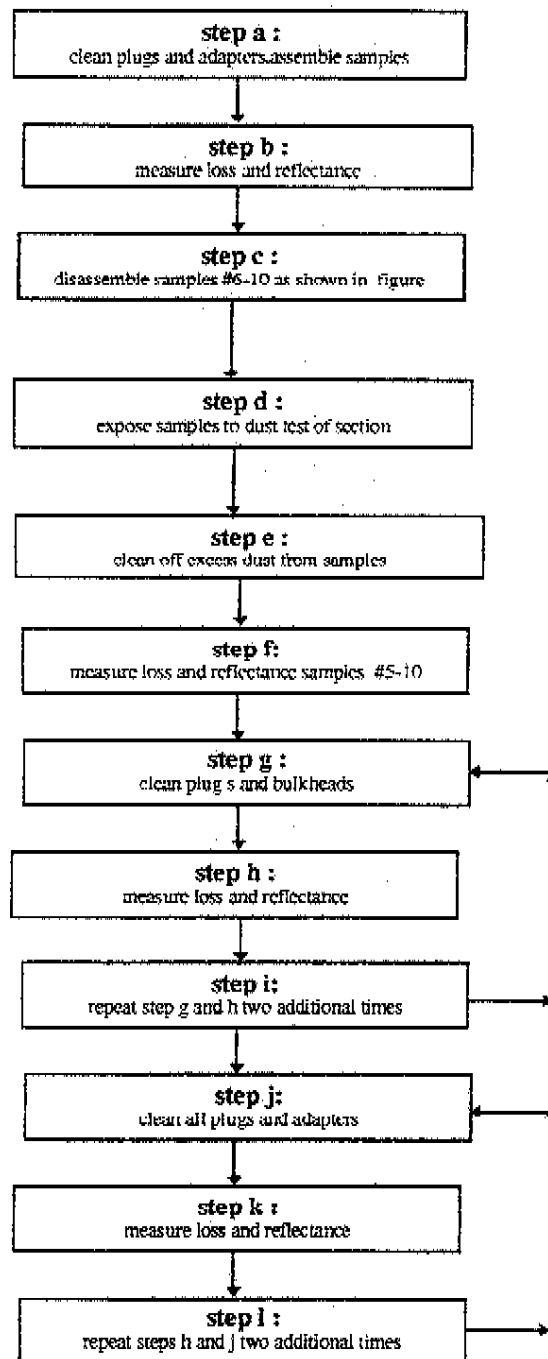
PRE
EXPOSUREPOST
EXPOSURE

Figure 4-6. Dust Test Flow Chart

**Single-Mode Optical Connectors and Jumper Assemblies
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... The criterion is **not met** if connectors which are non-conforming to the loss and reflectance Requirements criteria listed in Tables 4-2 and 4-3 after the first cleaning in Step "G" or Step "J" are brought back into conformance during the subsequent two cleanings.

RATIONALE — It is most desirable that the first cleaning be effective, and that subsequent cleanings are not required to bring the product into conformance.

O4-48 [66] Dust Resistance Objective

... The criterion is **not met** if product which is conforming after the first cleaning becomes non-conforming after a subsequent cleaning.

RATIONALE — It is most desirable that a product be resistant to degradation in performance from the very act of cleaning, since the recommended procedure is to clean after every demating.

4.4.4.2 Adhesive Testing

This test is designed to measure the strength of the adhesive bond between the fiber and the ferrule.

This test requires the use of a ceramic blank. This blank is to have an end face in the shape of a ferrule, but without a center hole. The exact dimensions and tolerances may be found in Section 5.3.6. The blank is then placed against the test plug so that the blank touches the ferrule as two plugs would in a connector assembly. A load equal to two pounds total is then applied to the plug under test via the blank.

The sample population required is five fresh connector plugs. It is not desirable to use plugs which had previously been exposed to Salt Spray or Airborne Contaminants, since any residue on the plug may damage the sensitive profilometer or interfere with interferometric measurements. Plug endface geometry is measured before and after loading.

R4-49 [67] Adhesive Test

... After subjecting the specimens to loading with a ceramic blank for 7 days at 65°C with uncontrolled humidity, the endface geometry shall still be within the tolerances allowed by the Fiber Extension Test (Section 4.4.5.1).

NOTE— As mentioned in Section 4.4.2.1 (Thermal Age Test) there is an overlap between the Adhesive Test and the Thermal Age test. If the Thermal Age Test is performed, then the duration of exposure for the Adhesive Test may be reduced to 24 hours.

5.3.7 Dust Test

Twenty Type I samples (plugs) are required. They are separated into the following groups which are exposed to the dust chamber separately. Although it may be possible to expose all samples simultaneously in a larger chamber, the specifications given here were developed for a given chamber volume, and would need to be adjusted to give equivalent exposure. Either five mated connector pairs from set A (Group I), five unmated connector ends from set B with adapters (Group II) or five unmated connector ends from set B without adapters (Group III) will be exposed at one time. The open end of the adapter for Group II is capped. The 5 unmated plugs of Group III will be capped. These Groups are illustrated in Figure 5-9.

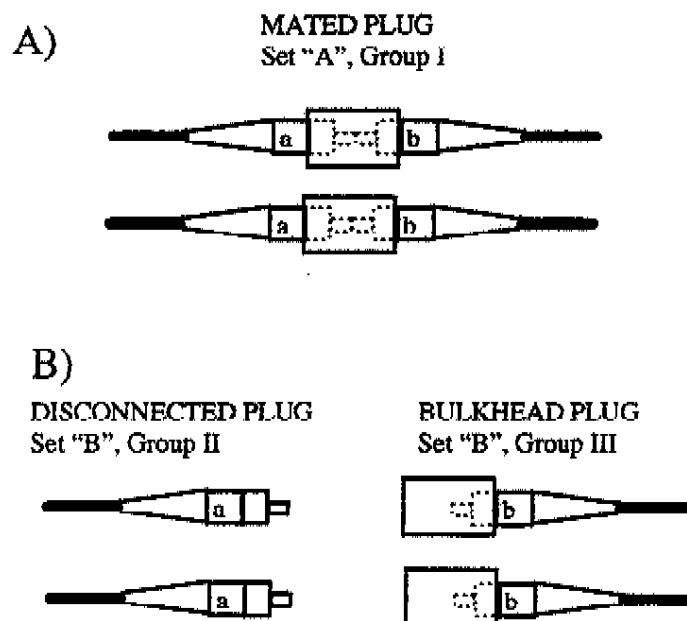
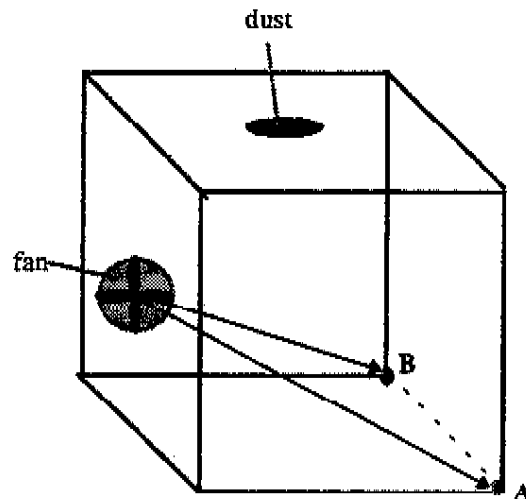


Figure 5-9. Dust Exposure Sample Groups

Samples are placed in a 12" by 12" desiccation box. Dust falls from the top of the box as an air source placed against one of the walls blows air into the far corners of the box to circulate the dust within the chamber. Figure 5-10 shows a schematic of the facility.

**Figure 5-10. Dust Test Facility**

The dust source will be "SAE Fine" or "Arizona Road Dust" [SAE J1669]. Tables 5-1 and 5-2 represent the particle size distribution and chemical composition of the dust to be used. Percentages are by weight.

Table 5-1. Particle Size Distribution

Particle Size μm	Weight %
0-5	39
5-10	18
10-20	16
20-40	18
40-80	9

Table 5-2. Chemical Composition

Component	Weight Percent
SiO_2	65-75
Al_2O_3	11-17
Fe_2O_3	2.5-5.0
Na_2O	2-4
CaO	2-5
MgO	1-2

Table 5-2. Chemical Composition (Continued)

Component	Weight Percent
TiO	0.5-1.0

This dust will be applied from the top of the desiccation box at a deposition rate of 8.5 ± 0.9 grams/min for two minutes. This will constitute one cycle of dust application. The deposition rate will be measured in still air.

An air source is centered horizontally with respect to the wall and vertically with its center approximately 3 inches above the box floor. The wind velocity of the air source should be 750 ft/min as measured outside the chamber in open space. The air source is positioned so that it blows the air into the far corners of the box, Point A and Point B in Figure 5-10. The air source alternates its direction from one corner to the next with each dust application cycle.

The samples will be exposed to 4 dust application cycles. All connectors and plugs should be positioned at least 1.5" from the side walls and floor, and 3" from the top of the box. The connectors should be spatially distributed throughout the box and may remain in the same position throughout the 4 cycles.

5.3.8 Groundwater Immersion Test

Five connections per medium shall be completely immersed in the five media described below for the Groundwater Immersion Test. An exposure of 7 days at a temperature of 22°C (72°F) $\pm 2^{\circ}\text{C}$ (4°F) is required. As a minimum the loss and reflectance shall be measured immediately upon immersion, then re-measured after 24 hours and at the end of the 7-day test period while the connections are still immersed.

1) Organism Exposure

Connections shall be immersed in a medium containing a mixture of the following organisms prepared from cultures containing an average of 1×10^8 organisms per milliliter per Table 5-3.

Table 5-3. Organism Exposure for Completed Splices

Organism	ATCC No.
Streptococcus salivarius	13419
Candida albicans	60193
Escherichia coli	11775
Penicillium aurantiogriseum	16025

4.2 Criteria

The general optical performance criteria are summarized in this section. The criteria for the specific tests are included with the respective tests.

4.2.1 Overview of Optical Performance Criteria

Optical measurements are to be made at both 1310 nm and 1550 nm for connectors with a single fiber joint. Criteria for the loss of hybrid splice/connectors with two or three sequential joints may be found in GR-2919-CORE. The loss of connectors with a single joint is only slightly dependent on wavelength, whereas the loss of connectors with multiple sequential joints is an oscillating function of wavelength and can vary by several tenths dB.

Tables 4-2 and 4-3 summarize the optical criteria. These criteria apply at both 1310nm and 1550 nm wavelengths. The specific numbered criteria are stated in Section 4.4.3, and refer back to these tables.

Table 4-2. Summary of Optical Performance Criteria: Loss

Test	Maximum Loss		Mean Loss		Loss Increase	
	(R)	(O)	(R)	(O)	(R)	(O)
New Product	0.50	0.20	0.20	0.15	---	---
During Test, Not Under Load	0.50	0.30	0.30	0.20	0.30	0.20
During Test, Under Load	---	---	---	---	0.50	0.30
End of Test	0.50	0.30	0.30	0.20	---	---

Table 4-3. Summary of Optical Performance Criteria: Reflectance

Test	Reflectance			Reflectance Change	
	(R)	(CR)	(CO)	(R)	(O)
New Product	-40	-55	-60	---	---
During Test, Not Under Load	-40	-55	-60	5	2

Table 4-3. Summary of Optical Performance Criteria: Reflectance
(Continued)

Test	Reflectance			Reflectance Change	
	(R)	(CR)	(CO)	(R)	(O)
During Test, Under Load	-40	-55	-60	5	2
End of Test	-40	-55	-60	5	2

IMPORTANT In the criteria which follow throughout this document, whenever reference is made to criteria found in Table 4-3, Conditional Requirement and Conditional Objective reflectance criteria should be used where appropriate to the product being tested.

The specific application of the criteria in Tables 4-2 and 4-3 is defined for each of the tests in the description of the respective test. The following, however, are general guidelines for the application of the optical performance criteria:

- The New Product criteria are applied to brand new product, as delivered by the supplier, product that has not been in service or subjected to any tests.
- The "During Test, Not Under Load" criteria are applied to measurements made during tests in which mechanical load is not applied.
- The "During Test, Under Load" criteria are applied to measurements made during tests in which mechanical load is applied.
- The End of Test criteria apply to measurements made at the end of a test with all stress removed and the product at normal ambient conditions.
- The Loss Increase Criteria apply to the difference between New Product and End of Test loss measurements.
- Note that the definition of "reflectance" is that of a negative number, as opposed to Optical Return Loss, which is a positive number. The terms "greater than" and "less than" are applied to the numerical values of the reflectance, and not the absolute values of these numbers. Thus, a -50dB reflectance is greater than a -60dB reflectance.

4.2.2 Modifications and Conditions

The following sections describe modifications and conditions which apply to the criteria stated in Section 4.4 and summarized in Tables 4-2 and 4-3.